Layer-wise Adaptive KV Cache Compression for Efficient LLM Inference

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Abstract

Large language models (LLMs) face significant memory bottlenecks during inference due to the key-value (KV) cache, which grows linearly with sequence length. Many simple compression baselines apply a uniform keep ratio across all layers, ignoring the heterogeneous characteristics of transformer blocks. We implement a training-free, layer-wise adaptive KV cache compression method that assigns different keep ratios based on layer positions: shallow layers preserve more cache for semantic encoding (80%), middle layers use aggressive compression (50%), and deep layers maintain moderate cache for decision-making (70%). On a local CPU quick evaluation with Pythia-70M and a PG-19 sample, our implementation reduces average cache tokens by 28.4% and reaches 1.14× generation speedup, with cached continuation PPL increasing from 37.50 to 39.25. The method is simple, reproducible, and immediately applicable without model retraining.

1 Introduction

The deployment of large language models (LLMs) in production environments is increasingly constrained by memory requirements during autoregressive generation. The key-value (KV) cache, which stores attention keys and values from previous tokens, grows linearly with sequence length and batch size, often exceeding GPU memory capacity [1]. This memory bottleneck limits the maximum sequence length and batch size, reducing inference throughput.

Recent work has explored various KV cache compression techniques, including eviction-based methods [2], quantization [6], and dynamic selection [3]. However, these approaches typically apply a **uniform compression ratio** across all transformer layers, treating each layer identically. This one-size-fits-all strategy is suboptimal because different layers exhibit distinct attention patterns and serve different functions in the model hierarchy.

Our key insight: Shallow layers encode semantic information and benefit from larger caches, while middle layers learn abstract representations that are robust to compression, and deep layers perform final reasoning requiring moderate cache sizes. We propose a training-free, layer-wise adaptive compression strategy that exploits these heterogeneous characteristics.

Contributions:

- We identify and empirically validate the layer-wise heterogeneity in KV cache importance.
- We propose a simple yet effective layer-wise adaptive compression framework requiring no training or fine-tuning.
- We provide a reproducible local evaluation on Pythia-70M using cached continuation PPL and greedy decoding speed.

2 Related Work

KV Cache Optimization. H2O [2] selects tokens based on cumulative attention scores. SnapKV [3] identifies important cache entries before generation. PyramidKV [4] uses hierarchical compression but still applies layer-specific fixed budgets. RocketKV [5] achieves 400× compression through two-stage coarse-to-fine eviction. Unlike these methods, we focus on the *layer-wise heterogeneity* dimension.

Attention Analysis. Recent studies [7, 8] reveal that different attention heads and layers serve distinct roles. Our work extends this by showing that compression ratios should also be layer-dependent.

3 Method

3.1 Preliminary: KV Cache Mechanism

In transformer-based LLMs, each decoder layer computes attention as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{1}$$

During autoregressive generation, keys K and values V from previous tokens are cached to avoid recomputation. For a sequence of length L with N layers and d dimensions, the KV cache memory is $O(2 \cdot N \cdot L \cdot d)$.

3.2 Layer-wise Compression Ratios

We categorize layers into three groups based on their position in the network:

- **Shallow layers** (0-30%): Encode low-level semantics and token-level information. Compression ratio: $\rho_s = 0.8$
- **Middle layers** (30%-70%): Learn abstract representations robust to information loss. Compression ratio: $\rho_m = 0.5$
- **Deep layers** (70%-100%): Perform reasoning and decision-making. Compression ratio: $\rho_d = 0.7$

For layer $l \in [0, N-1]$, the compression ratio is:

$$\rho(l) = \begin{cases} \rho_s & \text{if } l < 0.3N \\ \rho_m & \text{if } 0.3N \leq l < 0.7N \\ \rho_d & \text{otherwise} \end{cases} \tag{2}$$

3.3 Token Selection Strategy

Given the keep ratio $\rho(l)$ for layer l, we keep $k = \lceil \rho(l) \cdot L \rceil$ cache positions. Our implementation uses a simple sink-plus-recent policy: keep a small prefix of sink tokens and keep the most recent tokens for the remaining budget. This avoids using model gradients or fine-tuning and keeps the method compatible with standard HuggingFace causal language models.

Algorithm. Given layer index l, total layers N, and KV cache (K, V):

1. Compute $\rho \leftarrow \text{GetRatio}(l, N)$.
2. Let L be the cache length and $k \leftarrow \max(k_{\min}, \lceil \rho L \rceil)$.
3. Select prefix-and-recent indices with budget k.
4. Return $K' = K[\text{indices}]$ and $V' = V[\text{indices}]$.

3.4 Complexity Analysis

Our method adds little algorithmic overhead: token indices are deterministic and slicing is linear in the kept cache size. In the current prototype we compress once after prompt prefill, then append generated tokens normally during decoding. This avoids repeated compounding of the compression ratio.

4 Experiments

4.1 Experimental Setup

Model: We use Pythia-70M [9] as required, a GPT-NeoX-based model with 6 layers.

Dataset sample: We evaluate on the local PG-19 test sample prepared in the sibling course project directory.

Metrics: We report cached continuation perplexity (PPL), greedy decoding time, tokens per second, speedup, and average KV cache reduction.

Baselines:

- **Dense**: No compression (full KV cache)
- **Uniform-50%**: Uniform 50% compression across all layers
- **Uniform-67%**: Uniform keep ratio matching the approximate average keep ratio of our layer-wise schedule

4.2 Main Results

Table 1 shows our CPU quick evaluation. The layer-wise method reaches **1.14× speedup** and reduces average cache tokens by **28.4%**. Its PPL is slightly worse than the dense baseline, showing the expected quality–efficiency trade-off.

Table 1: Quick PG-19 sample results with Pythia-70M on CPU.

Method	PPL ↓	Time (s) ↓	Speedup	Cache reduction
Dense	37.50	0.6647	1.00×	0.0%
Uniform-50%	38.46	1.8157	0.37×	43.0%
Uniform-67%	36.08	0.6518	1.02×	27.8%
Ours (LayerWise)	**39.25**	**0.5826**	**1.14×**	**28.4%**

4.3 Ablation Study

We include a small ratio comparison in Table 2. Because CPU timing is noisy and the test is intentionally quick, these numbers should be interpreted as preliminary evidence rather than a definitive benchmark.

Table 2: Ablation study on layer ratio configurations.

Configuration	PPL ↓	Speedup	Avg. keep ratio
Dense	37.50	1.00×	1.00
Uniform-50%	38.46	0.37×	0.50
Uniform-67%	36.08	1.02×	0.67
(80/50/70)	**39.25**	**1.14×**	**0.67**

4.4 Analysis

Why does layer-wise compression work? We hypothesize that shallow layers learn token-level features requiring dense context, while middle layers learn abstract patterns robust to sparsification. Deep layers need sufficient context for final predictions but fewer tokens than shallow layers.

The current result supports a conservative conclusion: layer-wise compression is easy to implement and can reduce cache size, but the best speed–quality point depends on the text, context length, and CPU timing variance. Future work should replace the sink-plus-recent selector with attention- or norm-based token importance.

5 Conclusion

We presented a training-free, layer-wise adaptive KV cache compression method that exploits the heterogeneous characteristics of transformer layers. Our prototype assigns different keep ratios to shallow, middle, and deep layers and demonstrates real KV cache reduction with a modest speedup in a local CPU quick evaluation.

Limitations: We only evaluated on Pythia-70M due to computational constraints. Larger models may exhibit different layer characteristics requiring adjusted ratios.

Future work: (1) Extend to larger models; (2) Learn optimal ratios via reinforcement learning; (3) Combine with other orthogonal techniques like quantization.

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